SMART INDIA HACKATHON 2026

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

Teleportation-Based Quantum Digital Signature (QDS) Simulation, Statistical Verification, Attack Analysis and Auditable Evidence

SIH Problem Statement ID: SIH26141

Team: Egreen Quanta

Theme: Cybersecurity | Category: Software

Technical & Research Report

Project principle
Use quantum protocol operations and statistical decision rules for threat detection. Do not use AI/ML in the SIH detection pipeline. Report simulations and security claims separately.

Prepared for Smart India Hackathon 2026

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Report status: based on the SIH26141 problem statement, the supplied Quantum Malware Hunter README, and the terminal output pasted by the team on 10 October 2026. Reported run output is not presented as an independent re-execution.

1. Executive Snapshot

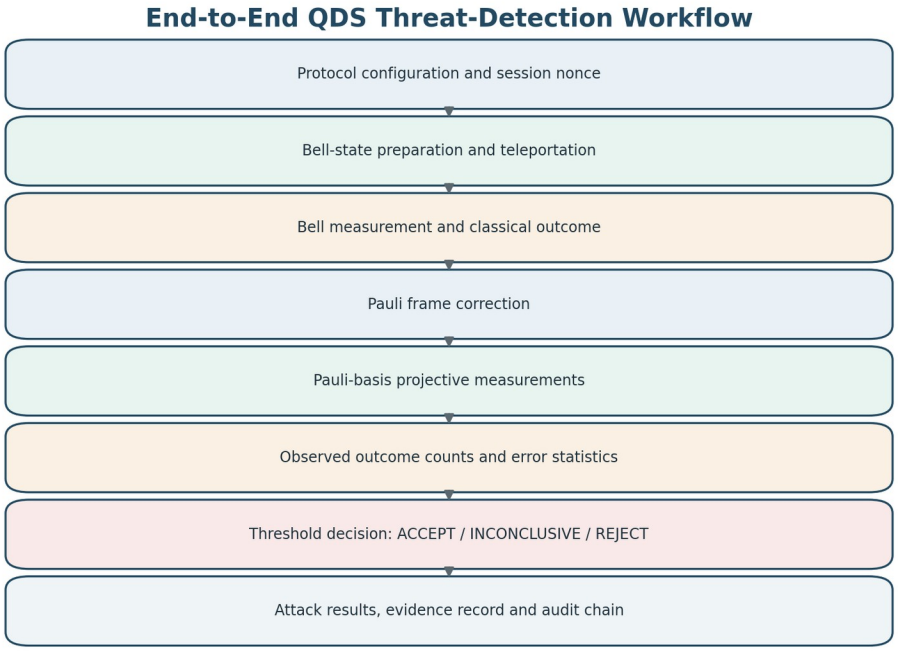


Figure 1. End-to-end workflow for the SIH26141 prototype.

SIH26141 asks for a quantum digital-signature threat-detection framework built around teleportation-based QDS. The framework is expected to simulate Bell-state entanglement and quantum teleportation, apply Pauli corrections and projective measurements, and analyse measurement outcomes using documented statistical rules to assess signature verification and attack behaviour [R1].

The current prototype run reported by the team has moved beyond the earlier ransomware-oriented demo output. It prints a protocol configuration, a single signature-verification trace, a Monte Carlo benchmark for five attack categories, a Chernoff-style statistical diagnostic, an audit-ledger integrity result, and a 20-test unit-suite summary.

The system design adapts the engineering pattern from the team's earlier Quantum Malware Hunter project: modular components, simulation-first operation, a digital-twin comparison, controlled threshold review, explainable evidence, and an audit trail. The SIH detection logic, however, must remain deterministic and protocol-based. The old Isolation Forest, autoencoder and SHAP components are not part of the SIH26141 target detection mechanism.

Core principle
A quantum signature verifier must report not just ACCEPT or REJECT, but the protocol configuration, measurement evidence, decision rule, and limitations behind that result.

2. Why This Problem

Digital signatures are used to authenticate a signatory and detect unauthorised changes to signed information. Quantum-capable adversaries could threaten some widely used classical public-key systems; this motivates work on quantum-resistant cryptography and on quantum cryptographic protocols. These are related but different approaches: post-quantum cryptography uses classical algorithms designed to resist quantum attacks, while QDS uses quantum information and protocol assumptions to provide signature properties [R6-R8].

The SIH26141 problem specifically focuses on threat detection for teleportation-based Quantum Digital Signature systems. Its stated attack categories include forgery, impersonation, replay, unauthorised verification, and quantum-channel manipulation. The requested method uses Pauli eigenstates, projective measurements, statistical analysis, and threshold-based decisions rather than AI/ML [R1].

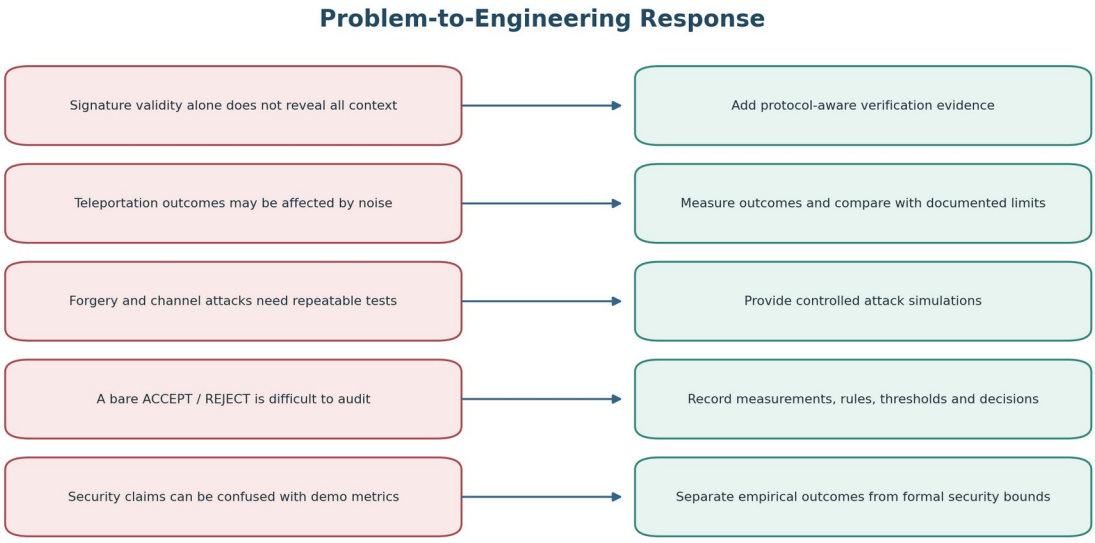


Figure 2. SIH problem areas mapped to engineering responses.

2.1 The gap being addressed

A signature verifier needs to distinguish an execution consistent with the protocol from one in which the evidence fails protocol-defined checks. At the same time, statistical outcomes can vary because of sampling and channel noise. A useful software framework must therefore expose the actual observations and the chosen threshold rule, and should not label every inconclusive trial as either a confirmed attack or a legitimate signature.

2.2 Scope boundary

- This report addresses a software simulation and analysis framework, not a claim of deployment over a physical quantum network.
- Replay rejection and participant authorisation may rely on classical session controls in addition to quantum measurements. The report identifies which check is responsible for each decision.
- A simulated attack result is evidence about the tested model and configuration. It is not by itself a formal proof of information-theoretic security.

3. Proposed Solution

The proposed SIH solution combines a documented QDS protocol model with a reproducible simulator, explicit verification rules, attack injection, per-trial telemetry, and an auditable evidence record. The framework generates legitimate runs and controlled attack scenarios, collects Bell measurement outcomes and post-correction measurements, calculates the observed error or deviation statistic, and applies configured thresholds to return an acceptance, rejection, or inconclusive result.

The current run reports a three-region threshold rule: ACCEPT when the observed error is at or below the verification threshold s_v; INCONCLUSIVE when the error is above s_v and at or below the dispute threshold s_a; and REJECT when the error exceeds s_a. The thresholds shown by the current configuration are s_v = 4.50% and s_a = 12.00%. These values are prototype settings, not universal constants for all QDS protocols.

3.1 Main capabilities

- Teleportation-based protocol simulation using three qubits per round, as reported by the current run output.
- Pauli-frame correction handling and projective measurements over configured Pauli bases X, Y and Z.
- Per-round match and mismatch accounting, empirical error rate and measurement-distribution statistics.
- Five controlled benchmark categories: legitimate baseline, signature forgery, impersonation, replay, and channel manipulation; unauthorised verification is included as the fifth attack category in the supplied table, while channel manipulation is the fifth when counting only attack classes. The benchmark in the supplied output includes all five attack classes plus the legitimate baseline.
- Threshold-based triage with an explicit inconclusive region for marginal results.
- Hash-chained audit records and automated unit tests.

4. System Architecture

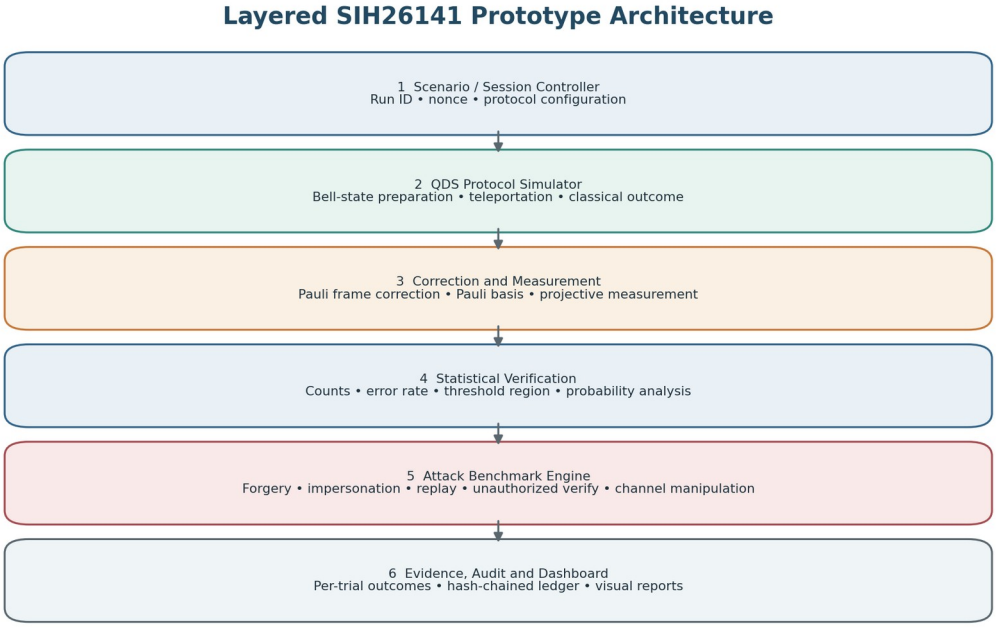


Figure 3. Logical layers of the SIH26141 system.

Layer	Responsibility	Evidence produced
Protocol configuration	Select variant label, round count, bases, noise and threshold values.	Version, config hash, seed and run ID.
Quantum protocol simulator	Prepare states, simulate Bell measurement / teleportation, apply correction.	State/circuit results and operation log.
Projective measurement	Sample outcomes in configured measurement bases.	Basis, outcomes and per-round matches.
Statistical verifier	Compute mismatch rate and apply decision regions.	Error rate, thresholds, decision rationale.
Attack benchmark	Run controlled scenarios and collect all trial outcomes.	Counts, detection rate, FAR/FRR, mean error.
Audit ledger	Store records and verify record-hash chain.	Entry IDs, hashes and integrity result.
Dashboard	Present actual simulator outputs and controls.	Charts, tables, run history and exports.

The layers should remain separate. In particular, the module deciding whether a QDS verification passes must not be replaced with the old machine-learning anomaly detectors from the ransomware prototype.

5. Workflow A: Teleportation-Based QDS Protocol

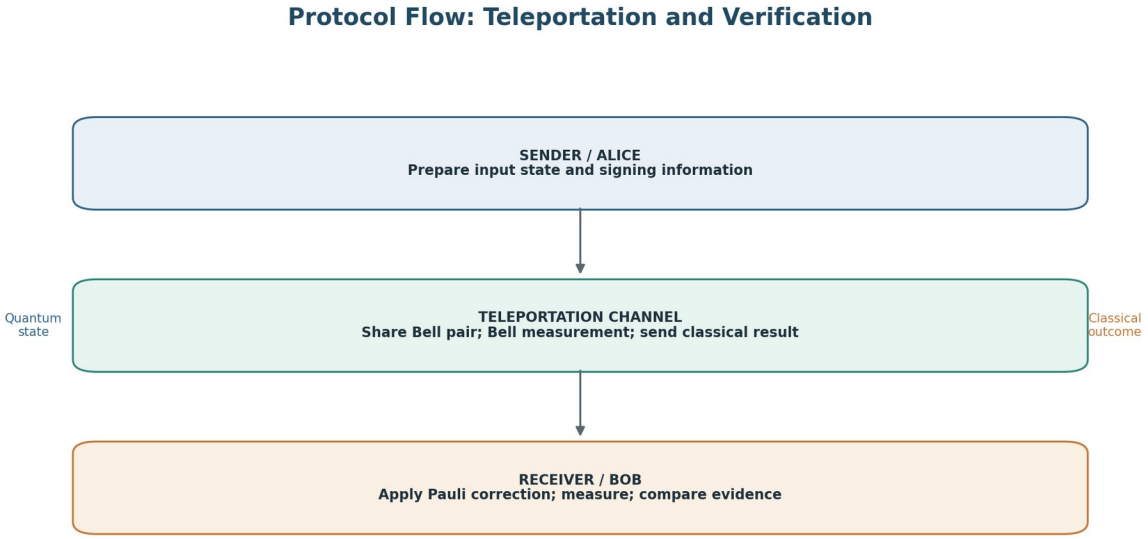


Figure 4. High-level teleportation and verification roles.

The supplied SIH problem statement expects the framework to model Bell-state entanglement, quantum teleportation, Pauli corrections, and projective measurement [R1]. In standard qubit teleportation, the sender and receiver share an entangled pair; the sender performs a Bell measurement on the input qubit and their share of the pair, communicates the classical measurement outcome, and the receiver applies the corresponding Pauli correction [R2, R4]. The exact way those operations form part of a signature protocol must follow the selected QDS construction, rather than treating teleportation alone as a signature scheme.

5.1 Current configuration reported by the prototype

Parameter	Value in supplied run	Interpretation
Protocol variant label	Teleportation-Based QDS (Collins-Lu Hybrid)	Configured code label. The exact mapping to a published primary protocol must be documented before claiming protocol equivalence.
Protocol version	v1.0.0	Prototype configuration version.
Signature elements / rounds	128 sequential rounds	Do not call this 128 qubits; output separately reports 3 qubits per round.
Circuit size	3 qubits per round	Reported simulation circuit size.
Measurement bases	Pauli Z, X, Y	Weights 0.5, 0.3, 0.2 in the supplied run.
Baseline noise e_0	1.50%	Configured depolarizing-noise baseline.
Verification threshold s_v	4.50%	Accept if observed error rate is at or below this value.
Dispute threshold s_a	12.00%	Reject if observed error rate is greater than this value; intermediate values are inconclusive.

5.2 Protocol verification note

The terminal reports successful execution of Bell-state teleportation, Pauli frame corrections and projective measurements. This report records those as outputs from the supplied run; it does not independently inspect the implementation source or reproduce the quantum calculations. Before the final submission, the team

should attach a specific source for the selected protocol and confirm that its message-signature relationship, participant roles, security parameter, and acceptance rules match the implementation.

6. Workflow B: Projective Measurement and Statistical Decisions

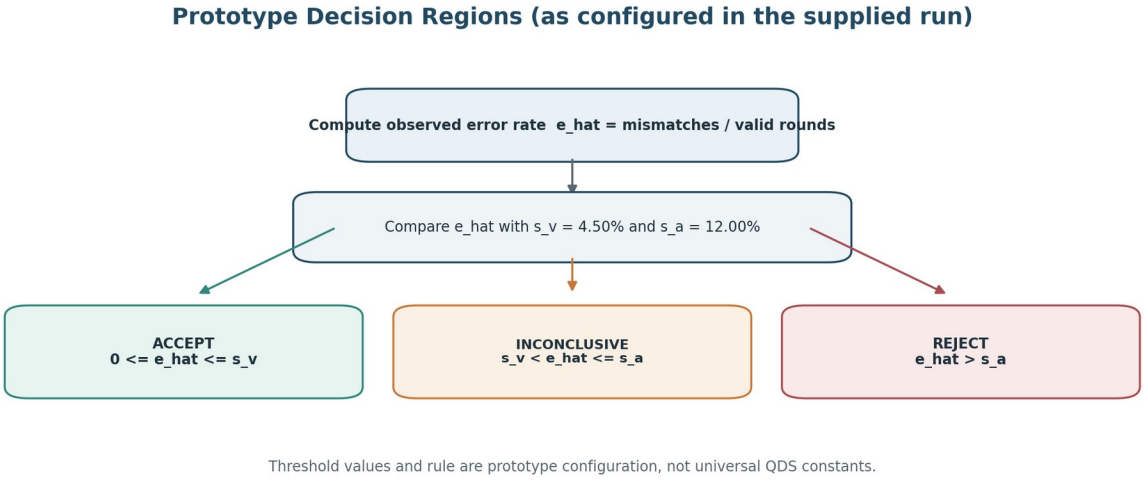


Figure 5. Three-region decision rule reported by the prototype configuration.

The framework derives an observed error rate from measurement comparisons. For N valid comparison rounds and M mismatches, the basic empirical mismatch rate is:

Observed error rate
e_hat = M / N. The formula is only meaningful when N > 0 and all counted rounds have well-defined comparison outcomes. Incomplete or invalid rounds must be recorded separately.

For the supplied configuration, the current prototype applies the following rule:

Condition	Prototype decision	Reporting requirement
0 <= e_hat <= 0.045	ACCEPT	Report mismatch count, round count, and rationale.
0.045 < e_hat <= 0.12	INCONCLUSIVE	Report that evidence lies between configured thresholds.
e_hat > 0.12	REJECT	Report threshold crossing and measurement evidence.

A real protocol may have other threshold semantics, and its values should be derived from the chosen security analysis and operating conditions. The thresholds above are the values reported by the current software, not a claim that 4.50% and 12.00% are generally secure for every QDS protocol.

6.1 Measurement-distribution and entropy telemetry

The run reports Shannon entropy $H = -\Sigma\ p_i \log_2(p_i)$ for an outcome distribution. The report should identify the exact distribution being measured (for example, the distribution of mismatch indicators or the distribution of measurement outcomes). Entropy is a descriptive statistic; it is not, by itself, a forgery probability or a proof of QDS security.

The run also reports a mean quantum-state fidelity of 0.9922 using $F = |\langle\psi_{in}|\psi_{out}\rangle|^2$. This value equals the displayed empirical agreement rate, 127/128 = 0.9922 after rounding. That equality does not automatically

make the value wrong, but the implementation should verify that fidelity is actually computed from a state overlap rather than copied from the agreement fraction.

7. Workflow C: Attack Simulation

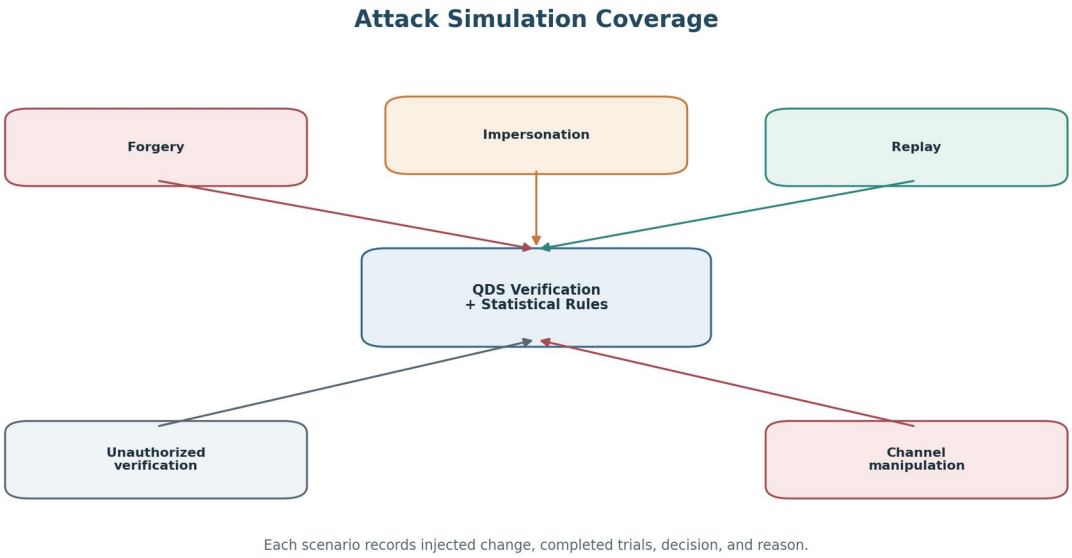


Figure 6. Attack classes routed through the deterministic verification pipeline.

The SIH problem asks for controlled detection of forgery, impersonation, replay, unauthorised verification, and quantum-channel manipulation [R1]. The prototype benchmark reports 30 Monte Carlo trials per category. Attack scenarios must alter the relevant protocol or session condition, and the result should indicate which rule detected the issue. For example, a replay may be stopped by a freshness nonce rather than by a projective measurement; that distinction matters in security analysis.

Scenario	What the test should exercise	Evidence to preserve
Signature forgery	Attempt to submit fabricated signature evidence under the selected QDS verification protocol.	Observed outcomes, mismatch count, protocol acceptance decision.
Impersonation	Attempt to act as an unauthorised signer or verifier.	Identity/session check result and any QDS measurement evidence.
Replay attack	Resubmit a previously valid session or transcript.	Nonce/session freshness decision and event identifiers.
Unauthorised verification	Attempt verification from an unauthorised participant/session.	Authorisation rule, identity binding and outcome.
Channel manipulation	Apply the configured quantum operation or noise manipulation to the simulated channel.	Applied manipulation, observed outcomes, error rate and decision.

A benchmark may show that every injected scenario was rejected or marked inconclusive under the tested setup. That is an observed result for those cases, not proof that all variants of each attack will always be detected.

8. Prototype Output and Benchmark Results

The following snapshot transcribes the terminal output supplied by the team for the run timestamped 2026-10-10 23:48:36. It is reported evidence from the team environment; it was not re-executed independently as part of preparing this report.

8.1 Single verification run

Metric	Reported result
Session ID	9150184cfc8c4b2c
Timestamp	2026-10-10T23:48:36
Total signature elements	128 measurement rounds
Projective matches / mismatches	127 / 128; 1 mismatch
Empirical agreement rate	99.22%
Observed error rate	0.78%
Mean quantum state fidelity	0.9922, reported formula $F = \|\langle\psi_in\|\psi_out\rangle\|^2$; requires source-code verification
Shannon entropy	0.0659 bits; reported outcome distribution needs to be documented
Pauli frame corrections	I: 33; X: 28; Z: 30; XZ: 37
Decision	ACCEPT, as e_hat = 0.0078 <= s_v = 0.0450
Audit block hash	9a1fe9c2a7629266d2a49b30b7b0319d0715deddd00ac1f5a573448bc2087c40

8.2 Reported Monte Carlo benchmark

Category	Trials	Accepted	Rejected	Inconclusive	Failed	Detection rate	FAR	FRR	Mean error
Legitimate baseline	30	30	0	0	0	N/A	N/A	0.0%	1.04%
Signature forgery	30	0	30	0	0	100.0%	0.0%	N/A	50.05%
Impersonation	30	0	30	0	0	100.0%	0.0%	N/A	50.05%
Replay attack	30	0	30	0	0	100.0%	0.0%	N/A	1.04%
Unauthorized verification	30	0	30	0	0	100.0%	0.0%	N/A	1.04%
Channel manipulation	30	0	11	19	0	100.0%	0.0%	N/A	11.69%

Metric rules shown by the implementation: attack detection rate = (rejected + inconclusive) / completed attack trials; FAR = incorrectly accepted attack trials / completed attack trials; FRR = rejected legitimate trials / completed legitimate trials. These definitions should remain explicit in the dashboard and source code.

Interpretation

The supplied run reports 0 accepted outcomes over 150 attack trials and 30/30 legitimate acceptances. For 150 independent attack trials with zero observed false acceptances, an approximate one-sided 95% binomial upper confidence limit is about 1.98%. This is a sampling statement under the tested conditions, not a cryptographic security guarantee.

8.3 Statistical diagnostics and audit

Reported output	Value	Interpretation / review requirement
Hypothesis null	H0: p <= 0.0150	Honest channel error hypothesis reported by code.
Alternative	H1: p >= 0.2500	Adversarial error hypothesis reported by code.
Standardized Z-score	-0.6690	Confirm exact statistic and tail convention in code.
P-value under H0	8.5551e-01	Document whether this is one-sided or two-sided and how it is calculated.
Chernoff FRR upper bound	7.8309e-02	Reported bound for P(e_hat >= s_v | H0); document derivation and assumptions.
“Security margin”	3.7 bits	Do not present as a formal cryptographic security level; it appears to be -log2 of the reported bound.
Audit ledger	5 hash-chained entries; VALID	Reported hash-chain check; this establishes

Reported output	Value	Interpretation / review requirement
		ledger tamper evidence, not QDS security.
Automated tests	20 passed; 0 failures/errors	As reported by user terminal output.
Streamlit dashboard	Did not launch: No module named streamlit	Dependency not installed in the current Python environment.

9. Mathematical Model and Security Interpretation

9.1 Empirical metrics

Let N be the number of completed and valid comparison rounds, M the number of mismatches, A the number of accepted attack trials, L the number of completed legitimate trials, and R_L the number of legitimate trials rejected. Then:

Core equations

Observed error rate: e_hat = M / N

Legitimate acceptance rate = accepted legitimate trials / completed legitimate trials

False rejection rate (FRR) = R_L / L

False acceptance rate (FAR) = accepted attack trials / completed attack trials

Attack detection rate (per the current benchmark) = (rejected + inconclusive attack trials) / completed attack trials

The implementation should report denominators beside percentages. It should avoid rounding small percentages into a misleading 0.0% where the exact count is more informative, and distinguish failed/incomplete simulations from protocol decisions.

9.2 Reported Chernoff bound

The supplied run reports p0 = 0.015, s_v = 0.045, and a Chernoff upper bound of 0.078309 for the probability that the observed error proportion crosses the verification threshold under the honest-channel assumption. A binary relative-entropy bound can be expressed as exp(-N D(s_v || p0)), where D(q || p) = q ln(q/p) + (1-q) ln((1-q)/(1-p)), for 0 < p < q < 1. With N = 128, p0 = 0.015 and q = 0.045, this expression is approximately 0.0783.

This is a statistical tail bound under the stated Bernoulli/error-rate assumptions. It is not the same as a formal forgery probability for a QDS protocol. Any claim of information-theoretic unforgeability requires the selected protocol's security theorem and threat model, not just the empirical error detector.

9.3 What the current run does not prove

- Thirty attack trials per class do not establish a universal 100% attack detection rate.
- The reported 3.7-bit value is not an independently established cryptographic security parameter and should be relabelled or removed from the main results until formally justified.
- The report does not claim that a hash-chained log proves signature authenticity; it demonstrates integrity checking over recorded audit entries.
- The exact protocol label "Collins-Lu Hybrid" appears in the configured output, but its primary-source mapping and equivalence to a published teleportation-based QDS variant must be documented before using it as a literature-backed protocol name.

10. Innovation and Uniqueness

The four ideas below adapt the engineering structure of the team's earlier Quantum Malware Hunter prototype. Its README describes entropy analytics, a digital-twin baseline, an adaptive cyber shield and explainable AI [R10]. The SIH26141 version retains the modular and demonstrable design pattern while replacing ML-based detection with protocol-specific quantum operations and deterministic statistics.

10.1 Innovation 1: Quantum Measurement Statistics

The framework records projective measurement outcomes, calculates error or deviation statistics, and compares results with protocol-defined or documented calibration thresholds. Entropy may be displayed as a descriptive view of outcome distributions, but it is not used as standalone proof of forgery.

10.2 Innovation 2: QDS Protocol Digital Twin

A controlled protocol model establishes an expected execution path. The simulator compares legitimate and manipulated runs so the team can inspect where a correction, measurement, session or channel condition changes the final verification evidence.

10.3 Innovation 3: Controlled Threshold Calibration

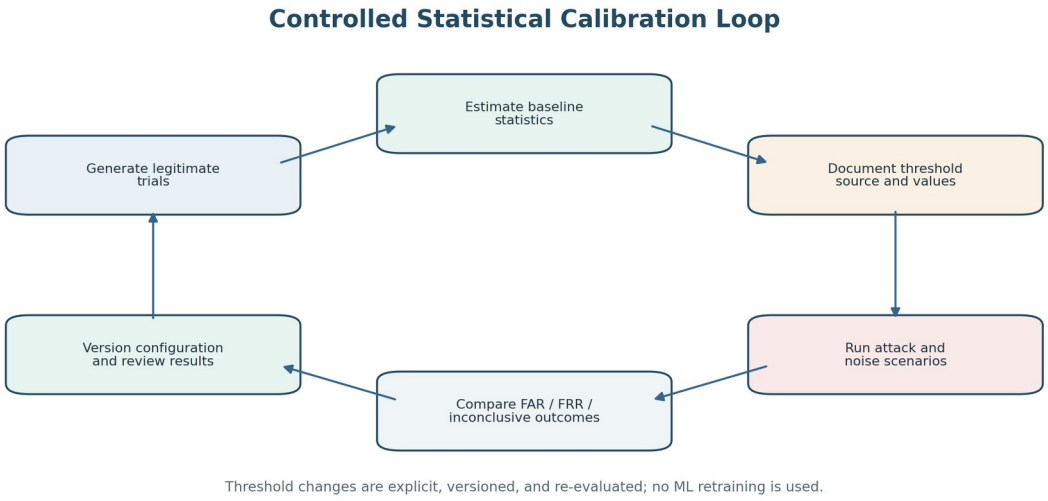


Figure 7. Explicit, versioned threshold calibration without machine learning.

Instead of incremental model training, threshold tuning is an explicit experiment. A configuration is selected, versioned, evaluated on legitimate and attack scenarios, and compared with earlier configurations. Changes are not triggered invisibly by incoming events.

10.4 Innovation 4: Explainable Verification Evidence

The system should explain decisions directly from deterministic inputs: measured counts, the selected basis, applied Pauli correction, decision threshold, statistical statistic and the specific rule that fired. The explanation is reproducible from the audit record and does not require an ML explainer.

10.5 Tamper-evident audit

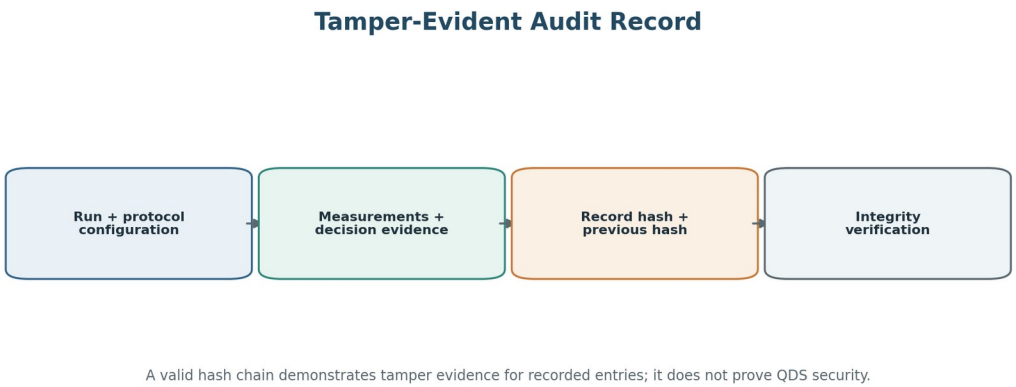


Figure 8. Audit record construction and hash-chain integrity check.

Hash-chaining audit records makes unauthorised modifications to stored entries detectable when the chain is verified correctly. It protects the history of software observations; it does not substitute for authentication, confidentiality, a QDS security proof, or a trusted quantum channel.

11. Technical Stack and Existing Prototype Mapping

Component	Current / suggested tool	Status from supplied evidence
Language	Python 3	Current CLI ran with python3.
Quantum simulation	Qiskit / local simulator	Prototype reports Bell-state teleportation and measurement execution; source-level validation still required.
Statistical analysis	Python / NumPy / SciPy candidate	Terminal reports error rates, entropy and Chernoff-style diagnostic.
Interactive dashboard	Streamlit	Dashboard startup failed in the shown environment because Streamlit is missing.
Audit	JSON hash-chained ledger	Five entries and integrity status reported in current run.
Tests	unittest	20 tests reported passing.
Configuration	protocol config with version/hash	v1.0.0 and hash prefix printed in current run.

11.1 Existing prototype vs SIH adaptation

Existing Quantum Malware Hunter capability	Reusable idea	SIH26141 adaptation	Status
Shannon / Rényi / other entropy analysis	Inspect outcome distributions	Use only as descriptive statistics for measurement outcomes; no standalone attack decision.	Existing concept; mapping requires validation
Quantum feature encoding + Isolation Forest	Modular simulator design	Replace with Bell-state / teleportation / Pauli / projective-measurement simulation.	Target pipeline must exclude ML
Digital-twin baseline	Compare expected and observed behaviour	QDS protocol execution twin and measurement-distribution comparison.	Prototype run reports comparison; implementation review needed
Adaptive retraining and model versions	Versioned feedback process	Controlled threshold/calibration versions; no ML training in the target detector.	Adapted design
SHAP and AI explanations	Readable explanation pane	Deterministic evidence summary generated from exact statistics and rule.	Replace AI/ML explanation path
Streamlit monitoring dashboard	Interactive display and charts	QDS verifier, attack lab, statistics, audit and run history.	Dependency missing at last run

Existing Quantum Malware Hunter capability	Reusable idea	SIH26141 adaptation	Status
Simulated / CSV data handling	Scenario and test data flexibility	Protocol input, attack injection, seed and reproducible result config.	Reusable infrastructure

12. Experimental Plan

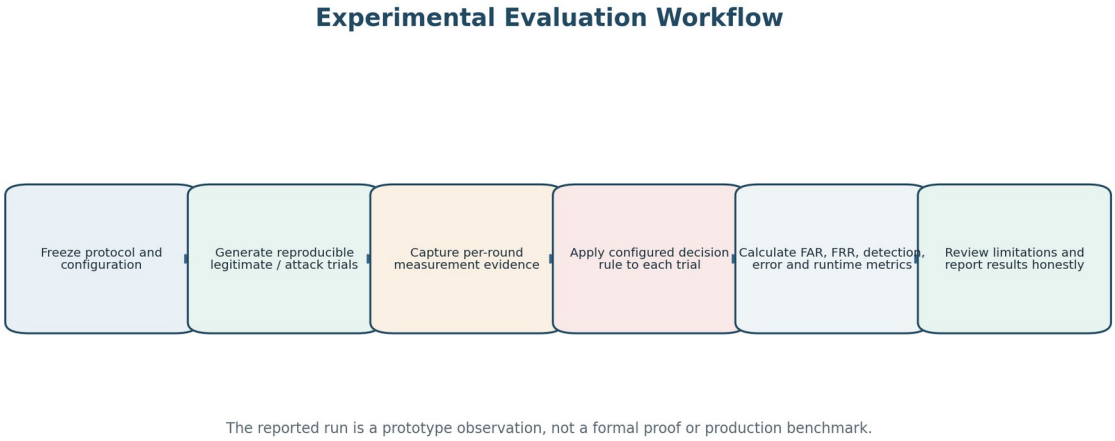


Figure 9. Reproducible evaluation workflow.

Experiment	Setup	Measurement / acceptance evidence	Status from supplied output
Single legitimate run	128 rounds; reported baseline noise 1.50%.	Matches, mismatches, e_hat, correction counts, decision.	Run reported
Legitimate baseline benchmark	30 trials.	Acceptance rate, FRR, mean error.	30 accepted reported
Signature forgery	30 trials.	Accepted / rejected / inconclusive, FAR, detection rate.	30 rejected reported
Impersonation	30 trials.	Identity/session check and protocol decision.	30 rejected reported
Replay	30 trials.	Freshness/session result and protocol decision.	30 rejected reported
Unauthorised verification	30 trials.	Authorization and verifier binding.	30 rejected reported
Channel manipulation	30 trials.	Noise/manipulation configuration and three-region decision.	11 rejected, 19 inconclusive reported
Unit tests	Test suite executed with unittest.	Pass/fail/error count.	20 passed reported
Dashboard integration	Launch dashboard entry point.	Dashboard imports and renders live backend values.	Blocked by missing Streamlit

The supplied benchmark output reports 30 trials for the legitimate class and 30 trials for each of five attack categories. If this is the complete set of the displayed benchmark, it represents 180 trials total: 30 legitimate and 150 attack trials. Keep fixed seeds and versioned protocol configurations so results can be repeated.

12.1 Required future experiments

- Test more random seeds and larger numbers of rounds; report confidence intervals for empirical rates.
- Check threshold sensitivity over documented noise levels, rather than reporting one fixed condition only.
- Use independent protocol test vectors to verify Pauli correction mappings and measurement outcome expectations.
- Document sample failure handling and ensure ACCEPT / REJECT / INCONCLUSIVE / FAILED counts always sum to the trial total.

- Record simulator backend, dependency versions, seed, protocol variant source, and complete run configuration for each experiment.

13. Feasibility, Risks and Mitigation

Risk / challenge	Why it matters	Mitigation
Protocol-source ambiguity	An implementation label alone does not prove equivalence to a published QDS protocol.	Select and cite one primary protocol; map each code stage to its actual protocol step.
Simulator abstraction	A local qubit simulator does not reproduce all physical channel/device effects.	State model assumptions and add only documented noise/measurement channels.
Threshold tuning	A threshold that accepts honest runs may also accept some attacks, and vice versa.	Report FAR, FRR, inconclusive outcomes and threshold provenance together.
Small test set	30 trials per attack class has limited statistical power.	Expand repeats and report confidence intervals.
Overstated security margin	A statistical bound is not automatically a QDS forgery theorem.	Separate empirical FAR, binomial confidence limits, statistical tail bounds and formal protocol security proofs.
Dashboard environment	Streamlit is missing in the shown interpreter environment.	Install the project requirements in an isolated virtual environment and test the documented launch command.
Audit interpretation	A valid hash chain only shows the stored record sequence is internally intact.	Avoid claims that audit integrity proves the cryptographic protocol secure.
AI/ML restriction	Legacy prototype components use ML and explainable AI.	Keep these out of the SIH26141 detection path; use deterministic protocol and statistical rules.

14. Security, Ethics and Research Boundaries

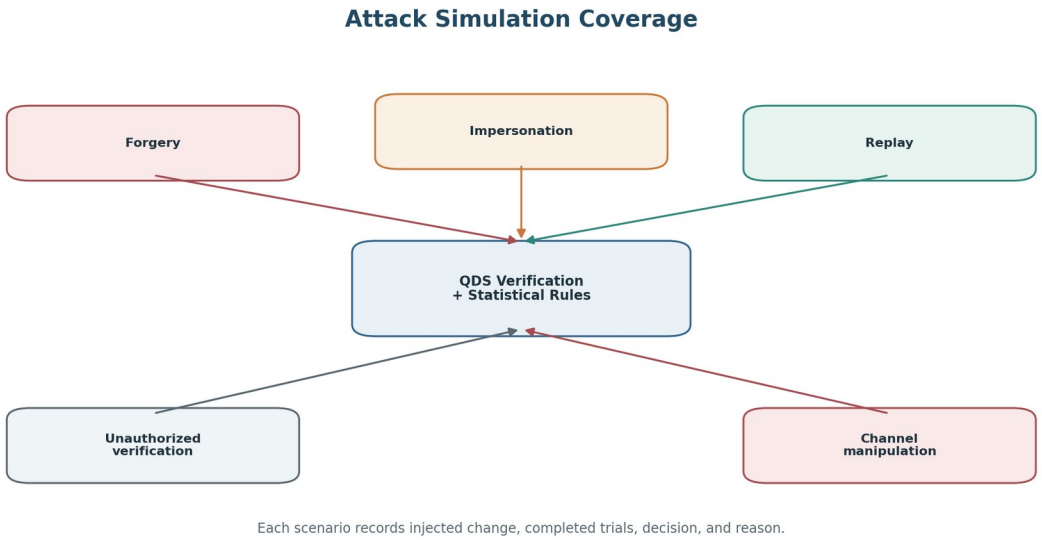


Figure 10. Attack scenarios tested through the local verification pipeline.

The report uses the phrase “information-theoretic security” only when tied to the assumptions and proof of a specific QDS protocol. QDS protocols may offer properties such as unforgeability, non-repudiation and transferability, but those properties depend on the protocol construction and threat model [R3, R5, R9]. A demo’s detection metrics cannot establish those guarantees independently.

- Do not collect secrets or represent them in dashboard screenshots. Synthetic keys, identities and sessions should be used in demonstrations.
- Make participant authentication and replay freshness explicit protocol/session checks where required.

- Clearly label simulated noise and attacker operations. A local simulated event is not a real quantum-channel attack.
- Store the configuration, seed and decision evidence behind each run so researchers can reproduce results.
- Do not report the current “3.7 bits” text as a security level. The supplied output suggests it is the negative base-2 logarithm of the displayed 0.078309 bound; its cryptographic meaning is not established by that transformation.
- Do not describe 0 false acceptances across the benchmark as proof of zero possible forgery probability.

15. Implementation Roadmap

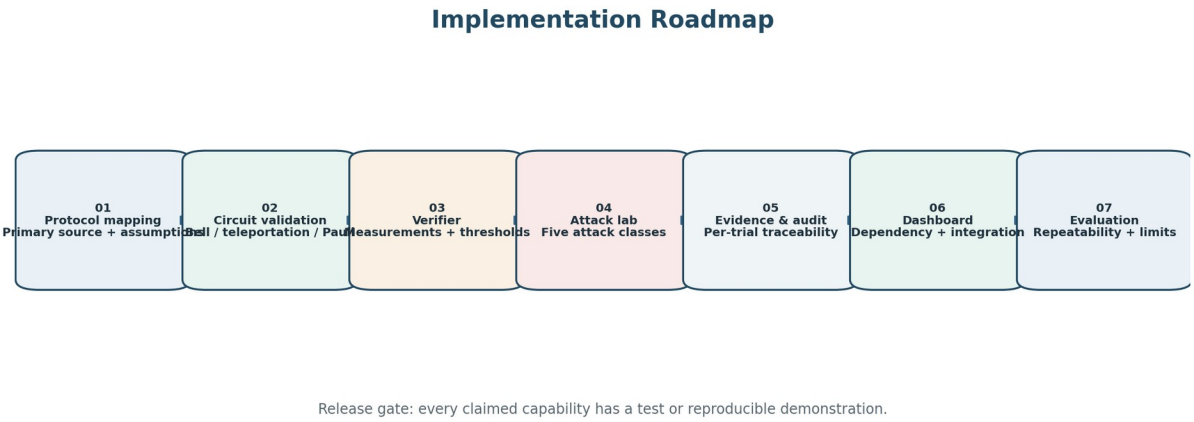


Figure 11. Suggested implementation and validation sequence.

Phase	Work	Exit criterion
1. Protocol selection	Cite exact QDS construction and list assumptions.	Each code stage mapped to a primary source.
2. Quantum operations	Verify Bell state, teleportation, measurement and Pauli corrections.	Unit tests against known state/circuit expectations.
3. Statistical verifier	Document threshold rules and edge-case handling.	Boundary tests and zero/insufficient sample tests pass.
4. Attack suite	Exercise forgery, impersonation, replay, unauthorised verification and channel manipulation.	Per-trial evidence and count invariants.
5. Audit ledger	Verify hashing and chain linkage.	Tampering test fails verification; normal chain passes.
6. Dashboard	Install dependency and connect all metrics to real results.	Dashboard launches and shows live run data.
7. Evaluation	Increase trials and test across noise/threshold settings.	Reproducible report with confidence intervals and limitations.
8. Documentation	Update README, usage, references and expected deliverables.	New team member can reproduce the demo.

16. Expected Impact

Audience	Expected value
Quantum cryptography researchers	A local environment for exploring QDS protocol executions, measurements and threshold behaviour.
Security researchers	Repeatable attack scenarios with visible decision evidence rather than opaque alerts.
Students and demonstrators	A clear interactive walkthrough of Bell-state teleportation and Pauli correction concepts.
Protocol developers	A place to compare noise settings, decision thresholds and

Audience	Expected value
	empirical verification rates.
Auditors / evaluators	Traceable experiment records, reproducible configuration and explicit limitations.

The immediate value is research and demonstration: make protocol behaviour and verification evidence easier to inspect. The supplied run is not evidence of field deployment, physical-channel protection, or production readiness.

17. Limitations and Future Scope

- The report relies on the team-supplied CLI output; the underlying code and complete test artifacts were not independently re-executed during authoring.
- The output prints “Collins-Lu Hybrid” as the configured protocol name. The exact primary-source match and protocol assumptions must be attached before the implementation is described as a faithful implementation of a published protocol.
- The single-run fidelity value equals the rounded empirical match fraction. Confirm the state overlap calculation in source code and test it independently.
- The P-value and Z-score are displayed but their exact test definition and tail convention need to be documented in code and README.
- The displayed Chernoff bound can describe a statistical tail event under specified assumptions, but it does not prove the complete QDS protocol’s forgery bound.
- The dashboard did not launch in the provided run because Streamlit was not installed in the active Python environment.
- The existing Quantum Malware Hunter ransomware prototype includes ML-based modules. Those modules do not satisfy the no-AI/ML constraint and must remain outside the SIH26141 threat-detection path.

17.1 Future scope

- Formal review of the selected QDS protocol and derivation of the exact security bounds it supports.
- More realistic noise, loss and device-error simulation where supported by protocol references.
- Expanded Monte Carlo analysis with confidence intervals and reproducible configuration export.
- Dashboard install/run validation and automated integration tests.
- Comparison with classical post-quantum digital signature standards as a separate context study, not as a substitute for QDS implementation [R6-R8].

18. Delivery Table (Expected Deliverables)

Expected deliverable	Description	Evidence / acceptance criterion	Current status
QDS protocol simulator	Bell-state and teleportation-based protocol simulation.	State/circuit tests and traceable end-to-end run.	Run reported; protocol mapping needs source verification
Bell-state and teleportation workflow	State preparation, Bell measurement, communicated result and correction.	Known-input tests and documented circuit steps.	Execution reported; independent source review needed
Pauli correction and projective measurement	Pauli-frame correction and X/Y/Z measurement outcomes.	Counts and operation trace; unit tests.	Run reports operations; tests reported passing
Statistical threshold verifier	Three-region decision logic with configured s_v and s_a.	Boundary tests and per-trial decision evidence.	Implemented behaviour reported; rule provenance to document
Attack-simulation module	Five attack classes against the deterministic verifier.	Trial counts, outcome reasons and metric calculations.	Benchmark run reported
Forging probability / security analysis	Statistical bounds and protocol-specific forgery analysis.	Derivation linked to a primary QDS protocol source.	Partial; formal QDS bound not established by run output
Performance evaluation report	FAR, FRR, detection, inconclusive rate, error, runtime and confidence intervals.	Reproducible test set and documented formulas.	Preliminary run metrics reported; expand trials
Interactive dashboard	QDS overview, simulator, attack lab,	Dashboard starts and uses backend	Not verified; Streamlit missing in

Expected deliverable	Description	Evidence / acceptance criterion	Current status
	statistics and audit views.	results.	last run
Audit ledger	Hash-chained run evidence and integrity validation.	Mutation/tamper tests and reproducible chain check.	Chain integrity and tests reported passing
Reproducible test configurations	Seed, protocol version, thresholds, simulator backend and dependencies.	README commands and saved run configuration.	Partial; consolidate documentation
Technical documentation	Architecture, assumptions, limitations, setup and expected outputs.	Team can reproduce the displayed run.	This report produced; README update remains needed

19. Conclusion

The SIH26141 prototype has the beginnings of a useful simulator-based QDS verification environment: the supplied run reports 128 measurement rounds, an explicit three-region decision, a five-category attack benchmark, a hash-chained audit ledger and 20 passing unit tests. The next step is to align every implementation detail to one fully cited teleportation-based QDS protocol, verify the fidelity and hypothesis-test calculations, and make the dashboard dependency reproducible.

The strongest submission will not be the one with the boldest security percentage. It will be the one that can trace every claim to a protocol rule, measurement count, equation, executed test, or primary-source reference. The framework should continue to use quantum-protocol operations and deterministic statistical verification rather than AI/ML detection, and should state exactly what is already implemented versus what still requires validation.

References

1. [R1] Smart India Hackathon 2026, SIH26141: Quantum-Inspired Cyber Threat Detection for Digital Signature Security. Official problem statement. https://sih2026.vuce.in/ps/SIH26141
2. [R2] Liao, C.-T., Bahrani, S., Ferreira da Silva, F., et al. “Benchmarking of quantum protocols.” Scientific Reports 12, 5298 (2022). https://doi.org/10.1038/s41598-022-08901-x
3. [R3] “Teleportation-based continuous-variable quantum digital signature.” Results in Physics 53 (2023), 107018. https://doi.org/10.1016/j.rinp.2023.107018. Note: this is a continuous-variable QDS construction; it should not be assumed identical to the prototype’s three-qubit model.
4. [R4] Bennett, C. H., Brassard, G., Crépeau, C., Jozsa, R., Peres, A., and Wootters, W. K. “Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels.” Physical Review Letters 70, 1895 (1993). https://doi.org/10.1103/PhysRevLett.70.1895
5. [R5] “A Verifiable Arbitrated Quantum Signature Scheme Based on Controlled Quantum Teleportation.” Peer-reviewed article available through PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC8774689/
6. [R6] National Institute of Standards and Technology (NIST). Digital Signature Standard (DSS), FIPS 186-5. https://csrc.nist.gov/pubs/fips/186-5/final
7. [R7] NIST. FIPS 204: Module-Lattice-Based Digital Signature Standard (ML-DSA), final (13 August 2024). https://csrc.nist.gov/pubs/fips/204/final
8. [R8] NIST. FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA), final (13 August 2024). https://csrc.nist.gov/pubs/fips/205/final
9. [R9] “Unconditionally Secure Quantum Signatures.” Entropy 17(8) (2015), 5635-5650. https://www.mdpi.com/1099-4300/17/8/5635
10. [R10] Team source material: “Quantum Malware Hunter” README and system summary supplied by Egreen Quanta. Used only to identify the existing architecture and reusable engineering concepts; not as proof of QDS-specific implementation.

Appendix A: Reproducing the Current Demonstration

The following commands were supplied by the team. The CLI and unit-test commands reportedly ran successfully in the team environment; the dashboard command failed because Streamlit was not installed in that Python interpreter.

CLI demonstration

```
python3 main.py
```

Unit tests

```
python3 -m unittest discover -s tests -p "test_*.py" -v
```

Dashboard command

```
python3 run_dashboard.py
```

Reported result: 20 tests passed, with zero failures and zero errors. Dashboard setup should use a project-local virtual environment and the repository requirements file. If Streamlit is absent from that file, add and pin it as a dependency before documenting the dashboard as runnable.

For reproducibility, include the Python version, Qiskit/simulator versions, dependency lock file, random seed, run configuration, and JSON audit export. Do not put secrets or real quantum-channel credentials in the demonstration package.

Appendix B: Implementation Status

Area	Evidence in supplied output	Status interpretation
Protocol CLI	Protocol configuration, teleportation/correction/measurement stages and verification telemetry printed.	Execution reported; mathematical/code review still required.
Statistical decisions	s_v/s_a region logic and per-class outputs shown.	Threshold path operational in supplied benchmark; source rules need citations.
Attack benchmark	Five attack categories, 30 trials each, outcomes and metrics shown.	Benchmark executed according to output; more trials needed.
Audit chain	5 hash-chained entries; status VALID.	Integrity check reported; protocol security is separate.
Unit suite	20 tests passed.	Test suite passed in supplied environment.
Dashboard	No module named streamlit.	Not runnable in last shown environment; dependency needs setup.
Formal security proof	Not shown by terminal output.	Not established by this report; link to chosen QDS proof is required.